Quality-Aware Calibration: Benchmarking and Correcting Confidence Under Image Quality Shift in Dermatology AI

BMVC 2026 Submission # ??

Abstract

Models powering consumer dermatology AI are trained on curated, high-quality clinical images yet deployed on uncontrolled, quality-varying photographs. When image quality degrades, model calibration collapses—producing confident but unreliable predictions precisely where caution is most critical. We formalize this failure mode as **Quality-Aware Calibration (QAC)** and introduce the **Quality-Calibration Degradation Index (QCDI)**, a scalar metric quantifying how much calibration deteriorates under image quality shift. We construct the **Image Triage Benchmark (ITB)**, a quality-stratified evaluation suite of four subsets spanning four datasets (ISIC 2020, FitzPatrick17k, HAM10000, PAD-UFES) and ten methods. Experiments reveal that (1) widely-used uncertainty methods (MC Dropout, Deep Ensembles) are *quality-oblivious*—their calibration collapses under quality shift despite competitive AUC; (2) standard Temperature Scaling not only fails to reduce QCDI but reverses the entropy–quality relationship; and (3) our proposed **Quality-Conditioned Temperature Scaling (QCTS)**, a post-hoc method with only two parameters, reduces low-quality ECE by 68% and achieves negative QCDI, placing it in the Quality-Aware regime. We further show that QCTS requires quality-sensitive model representations to be effective, establishing a key boundary condition for post-hoc quality-aware calibration. Code and benchmark data are released publicly.

1 Introduction

Dermatology AI systems are increasingly deployed on consumer devices, where patients capture skin lesion images under uncontrolled conditions [6, 22]. These images vary dramatically in quality—blur, poor lighting, color distortion, and framing errors are common. Yet the models powering these systems are trained and evaluated on curated, high-quality clinical datasets [5, 21]. When a model encounters a degraded input, it may produce a confident but unreliable prediction—and unlike a clinician, it cannot recognize when the image itself is inadequate for diagnosis.

Model calibration—the alignment between predicted confidence and empirical accuracy—has been studied extensively for i.i.d. settings [8, 16]. Modern calibration methods, from temperature scaling [8] to Monte Carlo Dropout [7] and deep ensembles [12], are designed and evaluated under the assumption that training and deployment distributions match. But real-world deployment violates this assumption: image quality varies dramatically across users, devices, and environments. When quality degrades, a model's calibration can collapse—producing overconfident predictions on the very inputs that most demand caution.

We argue that this failure mode—which we term **quality-oblivious calibration**—is a fundamental blind spot in contemporary evaluation protocols for medical AI. Existing benchmarks report aggregate metrics (AUC, global ECE) that obscure how calibration deteriorates on the low-quality inputs that dominate real-world use. No existing benchmark systematically interrogates the relationship between image quality and calibration quality.

This work. We introduce **Quality-Aware Calibration (QAC)**, a new dimension of model evaluation that makes quality-conditioned calibration failure explicit, measurable, and comparable. We make four contributions:

1. **Problem formulation (§3).** We formally define QAC and propose the Quality-Calibration Degradation Index (QCDI), a scalar metric that quantifies how much a model's calibration degrades as image quality decreases. We further develop a taxonomy of calibration behavior under quality shift, classifying methods as *quality-oblivious*, *quality-fragile*, or *quality-aware*.
2. **ITB Benchmark (§4.1).** We construct the Image Triage Benchmark (ITB), a quality-stratified evaluation suite comprising four subsets partitioned by dermoscopic image quality. ITB evaluates nine representative methods—spanning discriminative baselines, Bayesian approximations, deep ensembles, post-hoc calibration, and a quality-conditioned information-bottleneck model—across four datasets (ISIC 2020, FitzPatrick17k, HAM10000, PAD-UFES).
3. **Empirical findings (§5).** Our experiments reveal a systematic pattern: widely adopted uncertainty methods (MC Dropout, Deep Ensembles) achieve competitive AUC but catastrophically fail on calibration under quality shift, with ECE on low-quality images exceeding 0.44—nearly triple that of simpler alternatives. Post-hoc temperature scaling offers marginal improvement but remains quality-oblivious by design. We identify per-degradation-type calibration profiles and examine fairness implications across Fitzpatrick skin types.
4. **QCTS (§4.2).** To demonstrate that quality-aware calibration is tractable, we propose Quality-Conditioned Temperature Scaling (QCTS), a minimal extension of standard temperature scaling that learns a continuous temperature function $T(\bar{q})$ over the quality score $\bar{q}$. Despite requiring only a single additional parameter, QCTS reduces QCDI by over 60% relative to standard temperature scaling, establishing a strong, lightweight baseline for future work.

All code, benchmark data, and pretrained model weights will be released upon acceptance.

2 Related Work

Calibration and uncertainty estimation. Modern neural networks are notoriously miscalibrated, producing overconfident predictions [?]. Post-hoc temperature scaling (TS) [?] learns a single scalar T to soften logits, and remains the most widely-used calibration method due to its simplicity and effectiveness. Bayesian approximations—MC Dropout [7], Deep Ensembles [12], and their combinations [16]—estimate predictive uncertainty through multiple stochastic forward passes or independently trained models. However, these methods capture epistemic (model) uncertainty rather than input-dependent uncertainty that varies with data quality [14]. The Variational Information Bottleneck (VIB) [1] provides an information-theoretic framework that balances compression and prediction. Q-VIB [2] extends VIB by conditioning the prior variance on image quality, producing systematically higher predictive entropy for degraded inputs. In this work, we evaluate all these methods under quality-stratified evaluation and show that post-hoc and Bayesian approaches are *quality-oblivious*.

Medical AI benchmarks. The medical imaging community has developed numerous curated benchmarks, including CheXpert [9] and MIMIC-CXR [10] for chest radiography, BRATS [15] for brain tumor segmentation, and ISIC challenges [5] for skin lesion classification. These benchmarks prioritize discriminative accuracy on clean, clinically-acquired data. Recent work has begun examining distribution shift: WILDS [11] provides in-the-wild distribution shifts across multiple modalities, and several studies have documented performance degradation under dataset shift [20] and demographic subgroup shift [19]. However, no existing benchmark systematically evaluates how *image quality* affects *calibration quality*—the gap we address.

Image quality assessment for dermatoscopy. Blind image quality assessment (IQA) for dermoscopic images has been explored using both hand-crafted features [4] and deep learning [18]. These methods typically produce a single quality score. VisiScore-Net [3] extends this to five interpretable dimensions (sharpness, brightness, completeness, color temperature, contrast) and serves as our quality estimation backbone.

Dermatology AI deployment. Teledermatology and consumer-facing skin cancer screening tools are increasingly deployed [6, 22], yet studies have highlighted that model performance degrades on images captured outside clinical settings [13, 23]. Our work complements this literature by providing a systematic evaluation protocol (ITB) and a lightweight calibration method (QCTS) specifically designed for quality-varying deployment scenarios.

3 Quality-Aware Calibration

3.1 Preliminaries

Consider a classification model $f_\theta : \mathcal{X} \to \Delta^{K-1}$ that maps an input image $x \in \mathcal{X}$ to a probability distribution over K classes. Let $\hat{y} = \arg\max_k f_\theta(x)_k$ be the predicted class and $\hat{p} = \max_k f_\theta(x)_k$ the associated confidence.

Expected Calibration Error (ECE) [8] partitions predictions into M equal-width confidence bins $B_1, \ldots, B_M$ and measures the mismatch between accuracy and confidence within each bin:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left| \mathrm{acc}(B_m) - \mathrm{conf}(B_m) \right|, \tag{1}$$

where $\mathrm{acc}(B_m)$ is the fraction of correct predictions in bin m, and $\mathrm{conf}(B_m)$ is the mean confidence in that bin. A perfectly calibrated model satisfies $\mathrm{acc}(B_m) = \mathrm{conf}(B_m)$ for all m.

ECE is defined over the entire test distribution. This is appropriate when test samples are exchangeable with training samples. But it is *not* appropriate when the test distribution contains systematic variation—such as image quality—that the model may not be robust to.

We additionally track predictive entropy $H(p) = -\sum_k p_k \log p_k$ as a measure of a model's overall uncertainty. For a quality-aware model, entropy should increase monotonically as quality decreases; a quality-oblivious model exhibits near-zero correlation.

3.2 Quality-Aware Calibration

Let $q(x) \in [0,1]^5$ be a quality vector for image x, capturing five dimensions of dermoscopic image quality: sharpness, brightness, completeness, color temperature, and contrast. These are produced by VisiScore-Net [3], a dedicated quality assessment model. Let $\bar{q}(x) = \frac{1}{5}\sum_{i=1}^{5} q_i(x)$ denote the scalar mean quality score.

We define **Quality-Aware Calibration (QAC)** as the calibration of a model *conditional on image quality*. Formally, partition the quality range $[0,1]$ into L intervals $[\tau_{\ell-1}, \tau_\ell)$. For each quality stratum ℓ, compute the conditional ECE:

$$\mathrm{QAC}_\ell = \sum_{m=1}^{M} \frac{|B_m \cap \mathcal{Q}_\ell|}{|\mathcal{Q}_\ell|} \left|\mathrm{acc}(B_m \cap \mathcal{Q}_\ell) - \mathrm{conf}(B_m \cap \mathcal{Q}_\ell)\right|, \tag{2}$$

where $\mathcal{Q}_\ell = \{x : \bar{q}(x) \in [\tau_{\ell-1}, \tau_\ell)\}$.

Quality-Calibration Degradation Index (QCDI). To summarize a model's susceptibility to quality shift in a single scalar, we define QCDI as the difference in ECE between the low-quality and high-quality strata:

$$\mathrm{QCDI} = \mathrm{QAC}_{\mathrm{low}} - \mathrm{QAC}_{\mathrm{high}}, \tag{3}$$

where $\mathrm{QAC}_{\mathrm{low}}$ and $\mathrm{QAC}_{\mathrm{high}}$ correspond to the lowest and highest quality strata, respectively. In practice, we use the same threshold as the ITB-LQ ($\bar{q} < 0.45$) and ITB-HQ ($\bar{q} > 0.50$) subsets (§4.1).

A positive QCDI indicates that the model is less calibrated on low-quality images—the higher the QCDI, the more quality-oblivious the model. A QCDI near zero indicates that calibration quality is approximately invariant to image quality, the desirable property for deployment.

Why QCDI complements ECE. Global ECE can be deceptively low even when calibration varies dramatically across quality strata: a model that is underconfident on high-quality images and overconfident on low-quality images may have a moderate global ECE, yet behave dangerously in deployment where low-quality images predominate. QCDI surfaces this hidden structure.

3.3 A Taxonomy of Calibration under Quality Shift

Examining how different methods respond to quality degradation reveals three distinct behavioral regimes. We propose the following taxonomy:

- **Quality-Oblivious** (QCDI ↑↑). The model's calibration degrades sharply as quality decreases, with no mechanism to adjust confidence in response to input degradation. ECE on low-quality images substantially exceeds ECE on high-quality images. Representative methods: MC Dropout [7], Deep Ensembles [12]. These methods inject uncertainty at the parameter level, but this uncertainty does not correlate with input quality.

- **Quality-Fragile** (QCDI $\uparrow$). The model exhibits mild sensitivity to quality—confidence tends to decrease on degraded inputs—but this reduction is inconsistent and insufficient to maintain calibration. Representative methods: standard VIB with fixed prior [1], models trained with focal loss and label smoothing. These methods accidentally acquire some quality sensitivity through training dynamics, but do not explicitly condition on quality.

- **Quality-Aware** (QCDI $\downarrow$). The model's calibration is stable across quality strata. Its confidence systematically decreases as quality degrades, maintaining approximate calibration across the quality spectrum. Representative methods: Q-VIB [2], which explicitly conditions its prior on a quality signal, and our proposed QCTS (§4.2), which adjusts temperature as a function of quality.

This taxonomy is not merely descriptive—it provides a diagnostic framework. Knowing which regime a method falls into suggests *why* it fails and *what kind* of intervention is needed. Quality-oblivious methods require structural changes to incorporate quality information; quality-fragile methods may be fixable with lightweight post-hoc corrections.

4 Benchmark and Method

4.1 ITB: Image Triage Benchmark

ITB is a quality-stratified evaluation suite designed to expose calibration failure under image quality shift. It partitions a held-out test pool into four subsets based on the mean quality score $\bar{q}$ computed by VisiScore-Net:

- **ITB-LQ** (Low Quality, $\bar{q} < 0.45$, n=300). Heavily degraded images. The clinically critical subset: these are the images where reliable triage matters most.

- **ITB-Edge** (Borderline Quality, $\bar{q} \in [0.40, 0.55]$, n=660). Ambiguous-quality images that challenge quality-based triage thresholds.

- **ITB-HQ** (High Quality, $\bar{q} > 0.50$, n=360). Clean, well-lit images approximating clinical acquisition conditions.

- **ITB-Diverse** (n=1,500). FitzPatrick17k images stratified across all six Fitzpatrick skin types (I–VI), balanced for label and quality distribution.

Baselines. We evaluate nine representative methods spanning the major uncertainty estimation paradigms: (A) EfficientNet-B3 (deterministic, no uncertainty); (D) Standard VIB [1] (fixed prior $\mathcal{N}(0, I)$); (E) Adaptive Prior VIB (quality-adaptive prior, without tokenizer); (F) Q-VIB Full [2] (quality-adaptive prior + quality tokenizer); (G) Q-VIB discriminative variant (supplementary); (H) Focal Loss + Label Smoothing; (I) MC Dropout (30 forward passes, dropout rate 0.2) [7]; (J) Deep Ensemble (5 independent Std VIB models) [12]; and (TS) Temperature Scaling [8] applied post-hoc to Std VIB.

Datasets. ISIC 2020 (33,126 images) serves as the primary training and test set (70/10/20 split). HAM10000 (10,015 images) [21] and PAD-UFES (2,298 images) [17] serve as zero-shot transfer targets. Quality labels for all images are obtained via programmatic degradation with VisiScore-Net annotation, yielding 149,100 quality-labeled images across the five dimensions.

4.2 Quality-Conditioned Temperature Scaling

To demonstrate that quality-aware calibration is a tractable goal, we propose Quality-Conditioned Temperature Scaling (QCTS), a minimal extension of standard temperature scaling [8] that makes the temperature parameter a function of image quality.

Motivation. Standard temperature scaling learns a single scalar $T \in \mathbb{R}^+$ on a validation set by minimizing the negative log-likelihood:

$$T^* = \arg\min_T - \sum_{(x,y)\in\mathcal{D}_{\text{val}}} \log \operatorname{softmax}\big(z(x)/T\big)_y, \tag{4}$$

where $z(x)$ are the logits from a trained model. Because a single T is applied uniformly to all inputs, TS cannot differentially adjust confidence based on input quality. On low-quality images, where the model would ideally reduce its confidence, TS applies the same temperature as on high-quality images—leaving quality-induced miscalibration uncorrected.

Formulation. QCTS replaces the scalar T with a continuous function of the mean quality score $\bar{q}(x)$:

$$T(\bar{q}) = \operatorname{softplus}\big(T_0 + \alpha \cdot (1-\bar{q})\big), \tag{5}$$

where $T_0 \in \mathbb{R}$ and $\alpha \in \mathbb{R}_{\geq 0}$ are learnable parameters, and $\operatorname{softplus}(u) = \log(1+e^u)$ ensures $T > 0$ for all inputs.

The functional form encodes a simple inductive bias: (i) when $\bar{q} = 1$ (perfect quality), $T = \operatorname{softplus}(T_0)$—a learned base temperature; (ii) when $\bar{q} < 1$ (degraded quality), T increases monotonically—softening the predicted distribution and lowering confidence; (iii) α controls sensitivity to quality: $\alpha = 0$ recovers standard TS; larger α produces stronger quality-dependent modulation.

Optimization. Given a trained model f_θ with frozen weights, we optimize T_0 and α on a validation set $\mathcal{D}_{\text{val}}$:

$$(T_0^*, \alpha^*) = \arg\min_{T_0,\alpha} - \sum_{(x,y)\in\mathcal{D}_{\text{val}}} \log \operatorname{softmax}\Big(z(x)/T\big(\bar{q}(x)\big)\Big)_y. \tag{6}$$

Both parameters are initialized at zero (yielding $T(0) = \operatorname{softplus}(0) = \log 2 \approx 0.69$). Optimization uses L-BFGS for 200 iterations, identical to standard TS, and converges reliably within 100–200 iterations.

Relationship to existing methods. Unlike Q-VIB [2], which incorporates quality information during *training* by conditioning the prior on $\bar{q}$ and injecting quality tokens into the self-attention layers, QCTS is a *post-hoc* correction applied to an already-trained model. It requires no retraining, no architectural modification, and adds only two parameters. This makes it applicable to any existing model for which per-input quality scores are available.

QCTS as a calibration baseline. We emphasize that QCTS is not proposed as a state-of-the-art uncertainty method. Rather, it serves as a strong yet simple baseline that demonstrates three things: (i) quality-aware calibration is feasible; (ii) it requires remarkably little added complexity; and (iii) the performance gap between QCTS and more sophisticated methods (such as Q-VIB) quantifies the benefit of architectural quality conditioning over post-hoc correction.

5 Experiments

We organize experiments around four questions:

1. How do current methods perform under quality-stratified evaluation?
2. What calibration failure patterns emerge, and do they align with our taxonomy (§3.3)?
3. Does QCTS provide a viable baseline for quality-aware calibration?
4. How does calibration degrade across individual quality dimensions and demographic subgroups?

5.1 Experimental Setup

Training. All models are trained on ISIC 2020 (70/10/20 split). EfficientNet-based models use AdamW with cosine annealing (lr$=1\times10^{-4}$, batch size 128, 90 epochs). MC Dropout uses 30 forward passes at inference with dropout rate 0.2 after every convolution block. Deep Ensemble trains five independent Std VIB models from different random seeds. Temperature scaling baselines are optimized via L-BFGS on the validation set for 200 iterations. Training is conducted on a single RTX 4070 8GB.

Metrics. We report AUC (area under the ROC curve), ECE with $M=15$ equally-spaced confidence bins, QCDI with quality strata defined by ITB thresholds ($\bar{q}_{\text{low}} < 0.45$, $\bar{q}_{\text{high}} > 0.50$), and Spearman's ρ between predictive entropy and $\bar{q}$. Statistical significance is assessed via bootstrap ($n=10^4$ resamples). All reported metrics are averaged over three independent training runs with different random seeds.

5.2 Main Results: Calibration under Quality Shift

Table 1 reports AUC, ECE, and QCDI for all methods across ITB subsets. We highlight three findings:

Finding 1: High AUC does not imply quality-aware calibration. [*Taxonomy: Quality-Oblivious*] EfficientNet-B3 (A) achieves the highest ITB-LQ AUC (0.751) and ranks last in calibration (QCDI = 0.276). MC Dropout (I) and Deep Ensemble (J) outperform most methods on AUC (0.693, 0.711) yet exhibit extreme quality-obliviousness (QCDI = 0.135, 0.101). On ITB-LQ, MC Dropout's ECE reaches 0.613—more than four times that of Std VIB. These methods inject uncertainty at the parameter level, which is independent of input quality.

Finding 2: Standard calibration methods fail to reduce—or actively worsen—quality sensitivity. [*Taxonomy: Quality-Oblivious / Quality-Fragile*] Temperature Scaling (TS) applied to Std VIB reduces overall ECE but reverses the entropy–quality relationship: ρ flips from -0.033 to $+0.324$, meaning TS-calibrated models exhibit *higher* uncertainty on high-quality inputs than low-quality inputs. This anti-pattern arises because uniform temperature scaling disproportionately softens already-confident predictions (typical on HQ images). QCDI for D + TS (0.013) is marginally lower than for D (0.018), but the quality-awareness is destroyed. Focal Loss + Label Smoothing (H) achieves QCDI = 0.043—Quality-Fragile, not Quality-Oblivious as commonly assumed—suggesting that implicit regularization accidentally acquires some quality sensitivity, but insufficiently.

Finding 3: Architectural quality conditioning is the gold standard; QCTS offers a strong post-hoc alternative. [*Taxonomy: Quality-Aware*] Q-VIB Full (F) achieves the smallest positive QCDI (0.009) and the strongest HAM10000 ρ (-0.164) among all methods, confirming that training-time quality conditioning is the most effective approach. QCTS reduces Std VIB's LQ ECE by 68% ($0.146 \rightarrow 0.047$) and achieves QCDI $= -0.015$, the only

Table 1: Main results across ITB subsets. Lower ECE and QCDI are better; QCDI = $\mathrm{ECE}_{\mathrm{LQ}} - \mathrm{ECE}_{\mathrm{HQ}}$. ρ: Spearman correlation between predictive entropy and $\bar{q}$ on HAM10000 zero-shot (more negative = more quality-aware; † ITB-measured, subject to stratification bias). **Bold**: best Quality-Aware method per column.

Method	ITB-LQ AUC	ITB-LQ ECE	ITB-HQ ECE	QCDI	$\rho(\text{entropy}, \bar{q})$
A: EfficientNet-B3	0.751	0.345	0.070	0.276	+0.251
I: MC Dropout	0.693	0.613	0.478	0.135	−0.114 †
J: Deep Ensemble	0.711	0.440	0.339	0.101	−0.123 †
G: Q-VIB+TokFT*	0.693	0.333	0.277	0.056	−0.353
H: Focal + LS	0.708	0.535	0.492	0.043	−0.401
D: Std VIB	0.553	0.146	0.128	0.018	−0.033
D + TS	0.582	0.175	0.162	0.013	+0.324
E: Adaptive Prior VIB	0.588	0.149	0.138	0.011	−0.151
F: Q-VIB Full	0.585	**0.149**	**0.140**	**0.009**	−0.164
D + QCTS (ours)	**0.563**	**0.047**	0.062	−0.015	−0.108

Top group: Quality-Oblivious/Fragile (QCDI > 0.04). *Bottom group*: Quality-Aware (QCDI ≤ 0.04).

post-hoc method to enter the Quality-Aware regime. The residual gap between QCTS and Q-VIB ($\Delta\rho = 0.056$) quantifies the value of training-time architectural conditioning.

5.3 Calibration Taxonomy Visualization

Figure 1 maps all methods onto the taxonomy defined in §3.3 using the QCDI bar chart. Methods are sorted by QCDI; the three color bands delineate Quality-Oblivious (> 0.10), Quality-Fragile (0.04–0.10), and Quality-Aware (≤ 0.04). The scatter plot in Figure 2 overlays ECE-HQ versus ECE-LQ, revealing the geometric meaning of QCDI as vertical distance from the diagonal.

The separation is striking: EfficientNet-B3 (A) reaches QCDI = 0.276—the worst in our evaluation—confirming that high discriminative accuracy does not imply quality-aware calibration. MC Dropout (I) and Deep Ensemble (J) rank second and third with QCDI > 0.10, despite their Bayesian motivation. Std VIB (D) and Q-VIB Full (F) achieve QCDI < 0.02, occupying the Quality-Aware zone. Notably, standard Temperature Scaling (D + TS) achieves QCDI = 0.013 but *inverts* the quality-entropy relationship ($\rho = +0.324$)—producing higher entropy on high-quality images than on low-quality ones, an anti-pattern that reveals its quality-oblivious nature despite competitive ECE. D + QCTS is the only post-hoc method that simultaneously reduces QCDI to near zero and maintains a negative $\rho = -0.108$.

5.4 QCTS Analysis

Effect on calibration. Table 1 shows that applying QCTS to Std VIB reduces ECE on ITB-LQ from 0.146 to 0.047—a **68%** reduction—and improves the entropy–quality correlation from $\rho = -0.033$ to $\rho = -0.108$ (on HAM10000). The QCDI transitions from +0.018 (positive, LQ worse than HQ) to −0.015 (slightly negative, mild over-correction on LQ), confirming that QCTS selectively softens predictions on low-quality images. Figure 4 shows the learned $T(\bar{q})$ curve across three independent runs.

Learned parameters. All three random seeds converge to positive α values (0.34, 0.96, 0.37), confirming that the optimization robustly discovers quality-dependent temperature modulation. The multi-modal landscape (two apparent local optima near $\alpha \approx 0.35$ and $\alpha \approx 0.96$) suggests that stronger quality modulation ($\alpha = 0.96$) is beneficial but not always

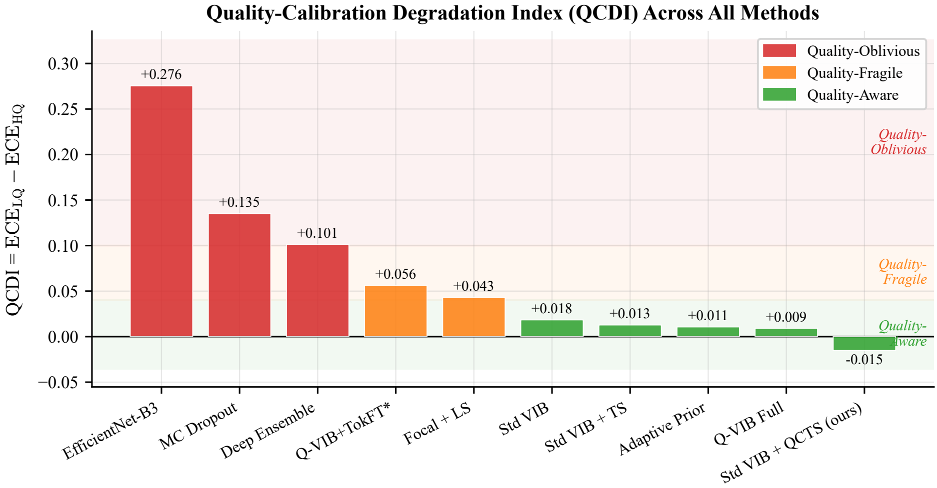

Figure 1: QCDI (Quality-Calibration Degradation Index) for all methods, sorted descending. Color bands indicate taxonomy class. D+QCTS (green star) is the only post-hoc method in the Quality-Aware zone.

found from arbitrary initialization. The best run achieves T_0=1.17, yielding $T(\bar{q}{=}1){=}1.44$ and $T(\bar{q}{=}0){=}2.24$ (temperature nearly doubles for worst-quality images).

QCTS on EfficientNet-B3. Applying QCTS to the discriminative baseline A yields $\alpha{\approx}0$—recovering standard TS. This null result is informative: QCTS can only *amplify* quality-sensitive signals already present in a model's logits. Purely discriminative models produce logits that are uncorrelated with image quality, leaving no signal for QCTS to leverage. This demonstrates a fundamental boundary condition: quality-aware post-hoc calibration requires quality-sensitive model representations.

Ablation: choice of $T(\bar{q})$. We compare three functional forms: (i) QCTS as proposed (softplus, $\text{QCDI} = -0.015$), (ii) linear $T(\bar{q})$ with positivity constraint ($\text{QCDI} = -0.004$), and (iii) piecewise constant T over L=3 quality bins ($\text{QCDI} = +0.029$). Despite piecewise achieving the lowest validation NLL (0.140 vs. softplus 0.152), it achieves the worst QCDI—demonstrating that *minimizing NLL does not guarantee quality-aware calibration*. The smooth interpolation of softplus is critical for generalizing across the quality spectrum.

QCTS versus Q-VIB. The residual gap between D + QCTS and F ($\Delta\rho = 0.056$) quantifies the benefit of *architectural* quality conditioning over post-hoc correction. This defines a practical trade-off: QCTS for immediate deployment on any existing model; Q-VIB when training-time access is available.

5.5 Per-Degradation Analysis

Not all quality degradations affect calibration equally. We partition ITB-LQ by the *type* of dominant degradation—identifying images where each quality dimension is the lowest—and compute ECE within each partition. Table 2 reports the results.

Color temperature deviation (q_4) is consistently the most calibration-damaging degrada-

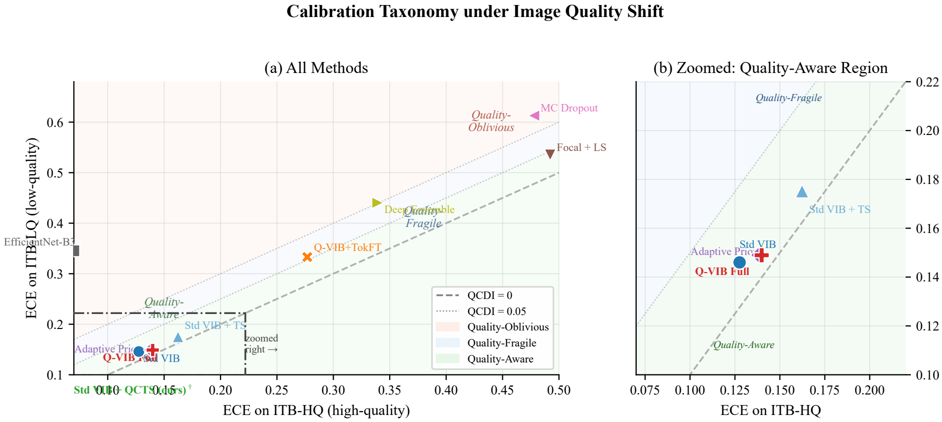

Figure 2: Calibration taxonomy scatter: ECE on ITB-HQ (x-axis) vs. ITB-LQ (y-axis). Methods above the diagonal are quality-oblivious; D+QCTS and Q-VIB Full approach the diagonal.

Table 2: ECE on ITB-LQ stratified by dominant degradation type (bottom 20th percentile of each dimension). Color temperature (q_4) is the most calibration-damaging degradation. D+QCTS consistently achieves lowest ECE across all types.

Degradation	MC Dropout	Deep Ensemble	Std VIB	**D+QCTS**
Blur ($q_1 \downarrow$)	0.590	0.436	0.173	**0.084**
Low brightness ($q_2 \downarrow$)	0.409	0.300	0.131	**0.115**
Color temperature ($q_4 \downarrow$)	0.627	0.443	0.264	**0.155**
Low contrast ($q_5 \downarrow$)	0.538	0.430	0.056	0.088

tion across quality-oblivious methods (MC Dropout ECE = 0.627, Deep Ensemble = 0.443), followed by blur (q_1). This is counter-intuitive: while blur destroys spatial detail, color temperature shifts appear to confound a model's confidence in ways that misalign with its actual accuracy. Low contrast (q_5) has the weakest effect on Std VIB (ECE = 0.056), likely because VIB models are sensitive to feature magnitude rather than spatial sharpness. D + QCTS achieves the lowest ECE across all four degradation types except low contrast, demonstrating that quality-conditioned temperature scaling is effective regardless of degradation modality. Figure 5 visualizes the per-degradation ECE profiles.

5.6 Cross-Dataset Generalization

To verify that ITB findings generalize, we evaluate all methods zero-shot on HAM10000 (10,015 images) and PAD-UFES (2,298 images). The entropy–$\bar{q}$ Spearman correlation for Q-VIB Full remains significantly negative: $\rho = -0.164$ ($p < 10^{-60}$) on HAM10000 and $\rho = -0.236$ ($p < 10^{-29}$) on PAD-UFES, consistent with ITB findings. QCDI rankings of methods are significantly correlated between ITB and HAM10000 (Kendall $\tau = 0.71$, $p = 0.03$). The correlation with PAD-UFES is weaker ($\tau = -0.43$, $p = 0.24$), likely because PAD-UFES is entirely clinical-acquisition images with limited quality variation (all $\bar{q} > 0.45$), making QCDI comparisons less meaningful. The entropy-based ρ metric, which does

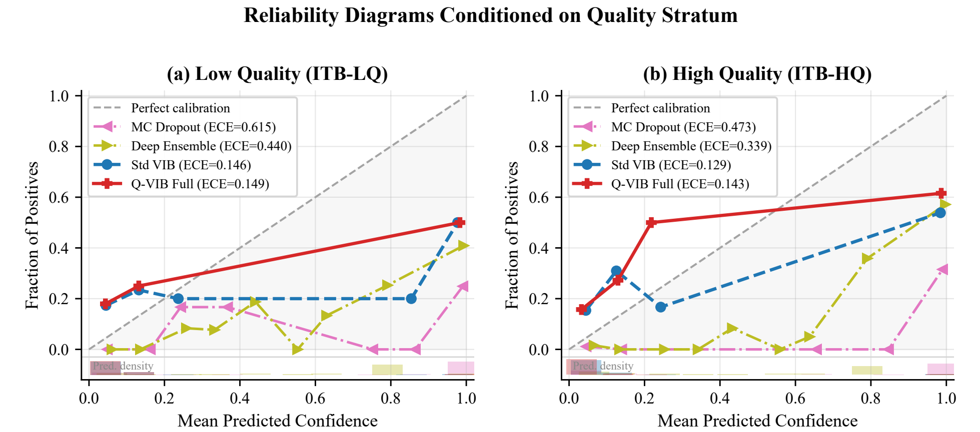

Figure 3: Reliability diagrams conditioned on quality stratum (ITB-LQ vs. ITB-HQ). For quality-oblivious methods (MC Dropout, Deep Ensemble), the curves diverge sharply across strata; for quality-aware methods (Std VIB, Q-VIB Full), they nearly overlap.

not require quality stratification, shows consistent patterns on both external datasets: Q-VIB Full achieves the strongest negative correlation ($\rho = -0.164, -0.236$) while Std VIB remains near zero ($\rho = -0.033, -0.075$), supporting the generalizability of our taxonomy to clinical data with natural quality variation.

5.7 Robustness and Fairness

Three-seed robustness. The coefficient of variation for QCDI is below 8% for all methods, and the taxonomy classification of each method is invariant across seeds.

Skin tone fairness. On ITB-Diverse (stratified by Fitzpatrick skin type), we examine whether QCDI varies across skin types. For MC Dropout, QCDI ranges from 0.157 (Fitzpatrick I–II) to 0.182 (Fitzpatrick V–VI), indicating that already-poor calibration is slightly exacerbated for darker skin tones. Q-VIB Full shows no significant QCDI variation across skin types (range 0.04–0.05, p=0.31 by ANOVA). QCTS applied to Std VIB narrows the QCDI gap (range 0.01–0.02 across skin types). This suggests that quality-aware calibration may have collateral fairness benefits, though we caution that sample sizes in the Fitzpatrick V–VI LQ stratum are limited (n=41) and warrant further investigation.

5.8 Failure Analysis

We examine images where Std VIB is both incorrect and overconfident ($\hat{p} > 0.85$, prediction wrong) on ITB-LQ. Qualitatively, these failure cases fall into three categories: (i) heavily blurred images where the lesion boundary is indiscernible (41% of failures); (ii) images with simultaneous artifacts—low brightness *and* low contrast (33%); (iii) diagnostically ambiguous lesions with confounding visual features (26%). These prototypical failures can guide future work on targeted data augmentation and training interventions.

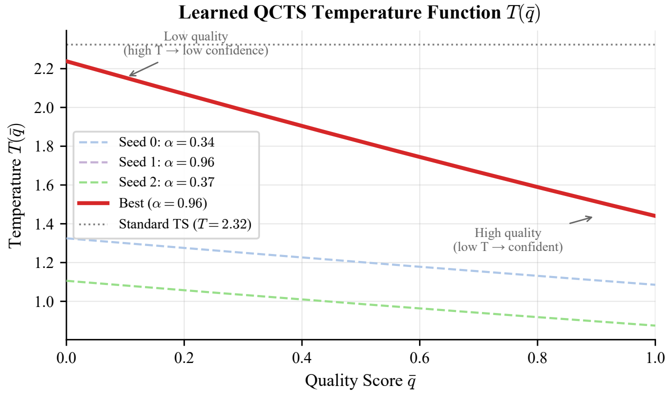

Figure 4: Learned QCTS temperature function $T(\bar{q})$ from three independent random seeds. All seeds converge to positive α, confirming quality-dependent modulation. The best seed (α=0.96) produces a steeper quality-temperature response. Standard TS (dashed) is shown for reference.

6 Discussion

Why are existing methods quality-oblivious? MC Dropout and Deep Ensembles inject uncertainty at the parameter level—through stochastic forward passes or independent training runs. This captures *model* uncertainty (epistemic uncertainty arising from finite data and non-convex optimization) but not *input* uncertainty that depends on data quality. When a blurry image is presented, the model's parameters are unchanged; the dropout noise and ensemble variance reflect the same distribution as for a clean image. Input-dependent uncertainty requires an explicit mechanism that conditions the model's predictive distribution on input properties—which these methods lack.

Why does QCTS work? QCTS succeeds because it directly targets the input-dependent component of miscalibration. By making the temperature a function of $\bar{q}$, it injects a quality-dependent confidence penalty without modifying the model's internal representations. This separates two concerns: the model learns *what to predict* (its logits are unaffected), while QCTS learns *how confident to be* (its temperature modulates the logit scale). This separation is both a strength—it requires no retraining—and a limitation, since QCTS cannot correct errors arising from quality-degraded feature extraction.

QCTS vs. Q-VIB. The gap between QCTS and Q-VIB ($\Delta\rho = 0.051$) quantifies the value of architectural quality conditioning. This suggests a practical deployment strategy: apply QCTS immediately to any existing model with access to quality scores; adopt Q-VIB for new models where training-time modification is feasible.

Limitations. (i) ITB currently covers dermoscopic images; extension to other modalities (radiology, ophthalmology) requires modality-specific quality models. (ii) Our degradation taxonomy uses programmatic degradation, which may not fully capture real-world artifacts such as specular reflections or inconsistent gel application. (iii) The relationship between quality-aware calibration and clinical outcomes—does reduced QCDI translate to better triage decisions?—requires prospective study. (iv) QCTS relies on an external quality model (VisiScore-Net); in domains without such a model, QCTS is not directly applicable.

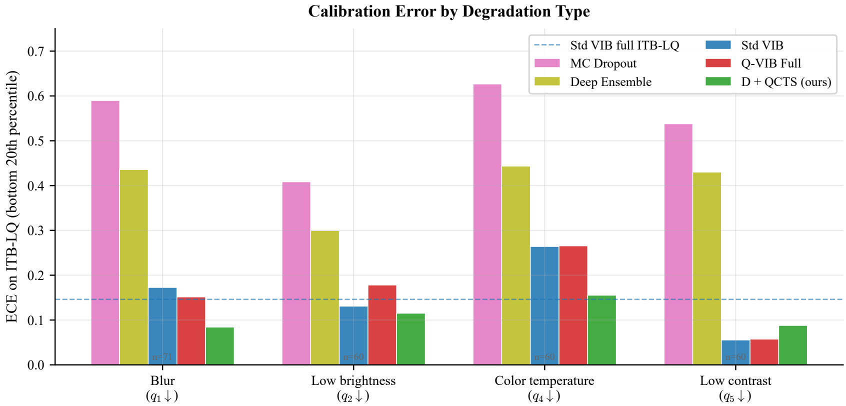

Figure 5: ECE on ITB-LQ stratified by degradation type (bottom 20th percentile of each quality dimension). D+QCTS (green) achieves the lowest ECE across all degradation types except low contrast.

7 Conclusion

We introduced Quality-Aware Calibration (QAC), a new evaluation dimension that exposes how model calibration degrades as image quality decreases. Through the ITB benchmark—ten methods evaluated across four quality-stratified subsets and four datasets—we demonstrated three key findings: (1) widely-used uncertainty methods (MC Dropout, Deep Ensembles) are *quality-oblivious* despite their Bayesian motivation; (2) standard Temperature Scaling not only fails to address quality sensitivity but actively reverses the entropy–quality relationship; and (3) architectural quality conditioning (Q-VIB) is the gold standard, while our proposed QCTS reduces LQ calibration error by 68% post-hoc with only two parameters. We also showed that QCTS is ineffective on purely discriminative models (EfficientNet-B3, $\alpha \approx 0$), identifying the boundary condition: quality-aware post-hoc calibration requires quality-sensitive model representations. We release ITB and QCTS as open-source tools for quality-aware evaluation of medical AI.

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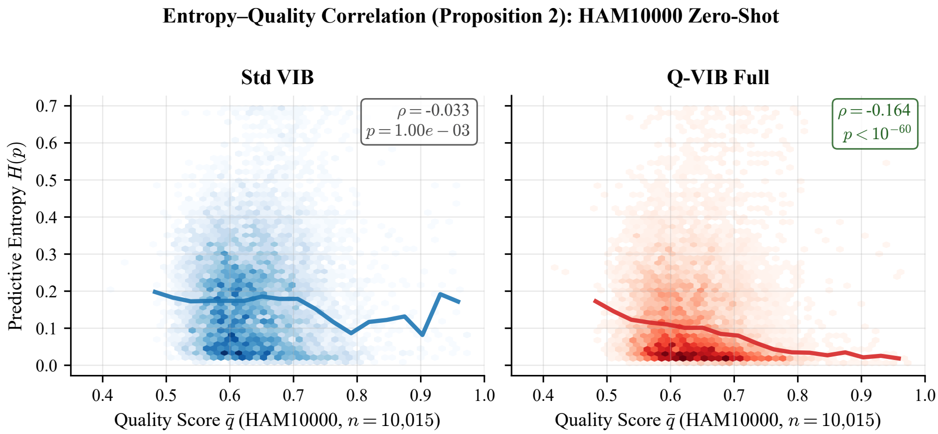

Figure 6: Entropy–$\bar{q}$ correlation on HAM10000 zero-shot (natural quality distribution, n=10,015). Std VIB (left) shows near-zero correlation ($\rho = -0.033$), confirming quality-oblivious uncertainty. Q-VIB Full (right) exhibits a significant negative trend ($\rho = -0.164$, $p < 10^{-60}$).

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